INFECTIOUS DISEASE MANUAL

CLINICAL STANDARD OPERATING PROCEDURE: INPATIENT COVID-19 MANAGEMENT DIRECTIVES

Respiratory Isolation Thresholds, Severity Risk Stratification, Antiviral/ Corticosteroid Inpatient Pathways, and Critical Surge Overrides

This policy blueprint establishes institutional parameters governing respiratory triage protocols, severe disease tracking frameworks, empiric antiviral or immunomodulatory dosing limits, emergency ventilatory escalation triggers, and negative-pressure zone overrides at St. Jude General Hospital. Optimized for technical parsing, indexing ingestion, and natural language retrieval in the enterprise AI Knowledge Ingestion Matrix.

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1. Introduction, Clinical Objectives, and Scope

Inpatient management of viral respiratory tract infections caused by severe acute respiratory syndrome coronavirus 2 requires strict adherence to standardized clinical parameters. Inconsistencies in risk assessment, delays in initiating early targeted antiviral therapy, or improper compliance with engineering controls increase the risk of healthcare-associated outbreaks, progressive hypoxemic respiratory failure, and systemic thromboembolic events.

The objective of this Standard Operating Procedure is to standardize respiratory care loops, isolation thresholds, and pharmacotherapeutic delivery across all adult services at St. Jude General Hospital. The protocol establishes cross-departmental standards to regulate patient flows from the point of initial intake through the critical care environment, minimizing diagnostic and barrier variance.

1.1 Scope and Clinical Enforcement

This mandate applies directly to all attending physicians, advanced practice providers (APPs), staff nurses, respiratory therapists, and clinical pharmacists across all physical infrastructure boundaries of St. Jude General Hospital. Compliance with all triage timelines and isolation rules is structurally mandatory.

2. Departmental Responsibility and Accountability Matrix

Maintaining safe containment boundaries while simultaneously deploying hyperacute respiratory therapies requires split, synchronized actions across hospital divisions:

- **Triage Nurse / Emergency Department Intake Specialist:** Responsible for immediate identification of matching upper/lower respiratory syndromic symptoms, enforcing source mask application, implementing immediate contact/airborne isolation containment, and executing rapid multiplex molecular screening draws.
- **Attending Physician / Hospitalist Provider:** Accountable for executing the baseline severity tier grouping, ordering radiographic mapping panels, checking laboratory hyper-inflammatory profiles, prescribing specified antiviral/immunomodulatory interventions, and documenting daily stability markers.
- **Respiratory Therapist (RT):** Tasked with continuous calibration of high-flow nasal oxygen (HFNO) loops, managing non-invasive or mechanical ventilation settings, logging real-time oxygenation trends, and coordinating safe manual proning maneuvers.
- **Clinical Pharmacist:** Tasked with verifying specific drug delivery timing rules, executing renal scaling dose reviews for antiviral compounds, cross-checking anticoagulant thresholds against hepatic or hematologic baselines, and monitoring drug-drug interactions.

3. Triage Risk Stratification and Severity Tier Placement

To preserve acute facility infrastructure and match asset distribution to patient risk profiles, all positive or suspected molecular panel cases must check into one of the following structural acuity tiers:

3.1 Clinical Stratification Boundaries

1. **Mild / Moderate Severity:** Patients with oxygen saturation profiles tracking at or above 94% on room air at sea level, showing no signs of progressive lower respiratory tract pathology or distress. These profiles are candidates for ambulatory out-of-hospital management plans.
2. **Severe Disease Status:** Clinically defined by a respiratory rate exceeding 30 breaths per minute, or an oxygen saturation profile tracking below 94% on ambient room air, or an active ratio of arterial oxygen partial pressure to fractional inspired oxygen (PaO_2/FiO_2) dropping below 300 mmHg. Admission to a dedicated respiratory medical ward is mandatory.
3. **Critical Severity Track:** Presenting with acute respiratory distress syndrome (ARDS), septic shock, metabolic encephalopathy, multi-organ breakdown, or progressive respiratory breakdown requiring advanced ventilatory mechanical intervention. Immediate critical care routing is required.

4. Step-by-Step Clinical Intake & Management Workflow

Upon laboratory verification of a positive molecular test result or highly presumptive imaging indicators, the primary team must follow this exact sequential workflow path:

Step 1: Immediate Source Barrier and Airborne Containment Enclosure

Apply a secure, well-fitted respirator mask to the patient immediately. Route the patient directly to a designated airborne infection isolation room (AIIR) configured with negative relative pressure infrastructure. Do not delay containment tracking for radiographic testing.

Step 2: Biomarker and Radiographic Baseline Mapping

Execute STAT laboratory profiling including a complete blood count, absolute lymphocyte metrics, liver function panels, serum creatinine, C-reactive protein (CRP), D-dimer, and ferritin. Obtain bedside chest X-rays or portable lung ultrasound maps to assess parenchymal infiltration metrics.

Step 3: Oxygen Target Calibration and Conservative Titration

Initiate supplemental oxygen via nasal cannula or high-flow interface to maintain oxygen saturation metrics strictly bounded between 92% and 96% for standard adult profiles, or strictly bounded between 88% and 92% for patients presenting with chronic hypercapnic respiratory diseases.

Step 4: Pharmacotherapeutic Loading Phase Deployment

Within the initial 24 hours of standard ward placement, order the specified institutional therapeutic package (Remdesivir, Dexamethasone, and antithrombotic profiling) matched directly to the patient's individual severity tracking code and filtration indices.

5. Pharmacotherapeutic Directives: Antiviral and Immunomodulatory Dosing

Inpatient medical ward therapies must adhere strictly to evidence-based dosing schedules adjusted cleanly for organ filtration profiles to eliminate facility toxicity trends:

5.1 Remdesivir (Antiviral Component) Delivery Rules

Indicated for hospitalized patients requiring supplemental oxygen within a strict 7-day window from symptom onset. Administer a single loading dose of 200 mg intravenously on Day 1, followed by a maintenance schedule of 100 mg intravenously daily for an absolute total duration of 5 days.

- **Organ Scale Limitation:** The clinical team must calculate baseline estimated glomerular filtration rate (eGFR). If a patient's eGFR drops below 30 mL/min, the standard compound formulation is contraindicated unless specified by nephrology directive.

5.2 Corticosteroid and Secondary Antithrombotic Formulations

- **Dexamethasone Regimen:** For all patients tracking within the Severe or Critical disease tiers requiring supplemental oxygen delivery, administer 6 mg of Dexamethasone either orally or intravenously daily for a max duration of 10 days or until discharge. Do not administer to mild cases.
- **Thromboembolic Prophylaxis:** Initiate standard prophylactic dosing using Low-Molecular-Weight Heparin (Enoxaparin 40 mg subcutaneously daily) within 12 hours of admission to mitigate high systemic hypercoagulability variables, unless active bleeding signs exist.

6. Exceptional Situations and Intensive Care Override Gateways

Sudden alterations in alveolar gas exchange parameters require an immediate bypass of ward therapeutic strategies to prevent cardiorespiratory arrest:

6.1 Progressive Respiratory Failure and Proning Triggers

If a patient tracking on high-flow nasal oxygen systems exhibits a respiratory frequency exceeding 35 breaths per minute, or shows persistent accessory muscle configuration use, or continues to track with an oxygen saturation level below 90% despite an inspired oxygen fraction (FiO_2) maximizing past 0.60, immediately coordinate a transition to Awake Self-Pruning. The patient must maintain a prone layout for a target benchmark of 12 to 16 hours daily.

> **CRITICAL SAFETY INTUBATION VALVE:** If the patient demonstrates severe clinical exhaustion, altered consciousness status, hemodynamic fragility, or a persistent oxygenation index (PaO_2/FiO_2) tracking below 150 mmHg despite non-invasive support optimization, activate the Emergency Intubation and Airway Team directly. Transition to protective invasive mechanical ventilation.

7. Quality Assurance Framework and AI Retrieval Ingestion Anchors

To secure system certification standards and limit facility cross-transmission indexes, the Data Quality Council reviews randomly extracted infectious disease electronic charts weekly.

7.1 Core Performance Metrics Checklist

- **Time-to-AIIR Isolation:** Target threshold requires ≥ 98% of suspected respiratory infectious profiles to achieve proper negative-pressure containment layout enclosure within 30 minutes of arrival.
- **Stewardship Compliance Latency:** Review and stop duplicate empiric antibiotic orders within 48 hours unless a secondary bacterial superinfection is laboratory-verified.

8. Automated AI Knowledge Assistant Search Queries

To facilitate instant structural parsing and natural language querying of specific manual directives by medical providers via interface terminals, the following technical query anchor codes are mapped directly to this text layer:

- `Query: [SOP-COV-STRATIFICATION]` Extracts the specific criteria parameters governing Mild, Moderate, Severe, and Critical care boundaries.
- `Query: [SOP-COV-PHARMACOLOGY]` Pulls up the complete dosing parameters, timing restrictions, and eGFR renal clearance limitations for Remdesivir and Dexamethasone loops.
- `Query: [SOP-COV-PRONING]` Accesses real-time execution steps, contraindications, and monitoring charts for the daily awake proning sequence.